

Technical Report: Latent Diffusion Model for Targeted Artifact Correction in MEG

1. Introduction and Objectives

Artifact correction in magnetoencephalography (MEG) signals traditionally relies on spatial projection methods such as Independent Component Analysis (ICA) or Signal Space Projection (SSP). These global approaches have a major limitation: they alter the temporal structure of the signal beyond the corrupted segments, potentially modifying clean signal areas. This project introduces a generative approach aiming to exclusively target time windows containing an artifact, reconstructing a signal whose local temporal dynamics are consistent with adjacent segments.

2. Model Architecture

The approach relies on a two-stage conditioned latent diffusion model, combining a quantized autoencoder for feature extraction and a diffusion model based on a Diffusion Transformer (DiT).

- **Vector Quantized Variational Autoencoder (VQ-VAE):** Pre-trained on artifact-free signal segments, it projects the time series into a discrete latent space. The latent codebook consists of 128 vectors of dimension 64. The model applies a temporal compression factor of 4, thereby reducing the dimensionality of the sequence to be processed by the underlying diffusion model.
- **Diffusion Model (Flow Matching):** The generative process uses an architecture inspired by *Text-to-Series (T2S)* (Ge et al., 2025). The approach is based on a Flow Matching (Rectified Flow) framework, which models an optimal transport trajectory between noise and latent data (modeled by an ordinary differential equation, ODE).
- **Denoiser and Conditioning:** The noise estimator is a Diffusion Transformer (DiT). Unlike text conditioning, the model is conditioned here by a vector of 22 features extracted via *catch22*. This conditioning is injected into the network through an Adaptive Layer Normalization (AdaLN) layer, which dynamically modulates the Transformer's normalization parameters based on the feature vector.

3. Training Setup

The training adopts a self-supervised paradigm using exclusively artifact-free signals. Performance tracking relies on several metrics: Fréchet Inception Distance (FID), Power Spectral Density (PSD) distance, latent space standard deviation, and feature space distance.

- **Diffusion Process:** The sampling of time step t follows a learning curriculum. During the first 5 epochs, an increasing uniform sampling is applied. The process then switches to a Beta(1.0, 0.8) distribution to optimize learning on the relevant dynamics.
- **Loss Function:** The optimization strictly minimizes the Mean Squared Error (MSE) between the network's predicted velocity and the target velocity of the optimal transport trajectory.

- **Optimization:** Training is performed using the AdamW optimizer (fixed learning rate of $1e-4$, weight decay of $1e-4$). The training strategy maintains a constant rate without relying on a learning rate scheduler.

4. Inference, Dataset, and Preprocessing

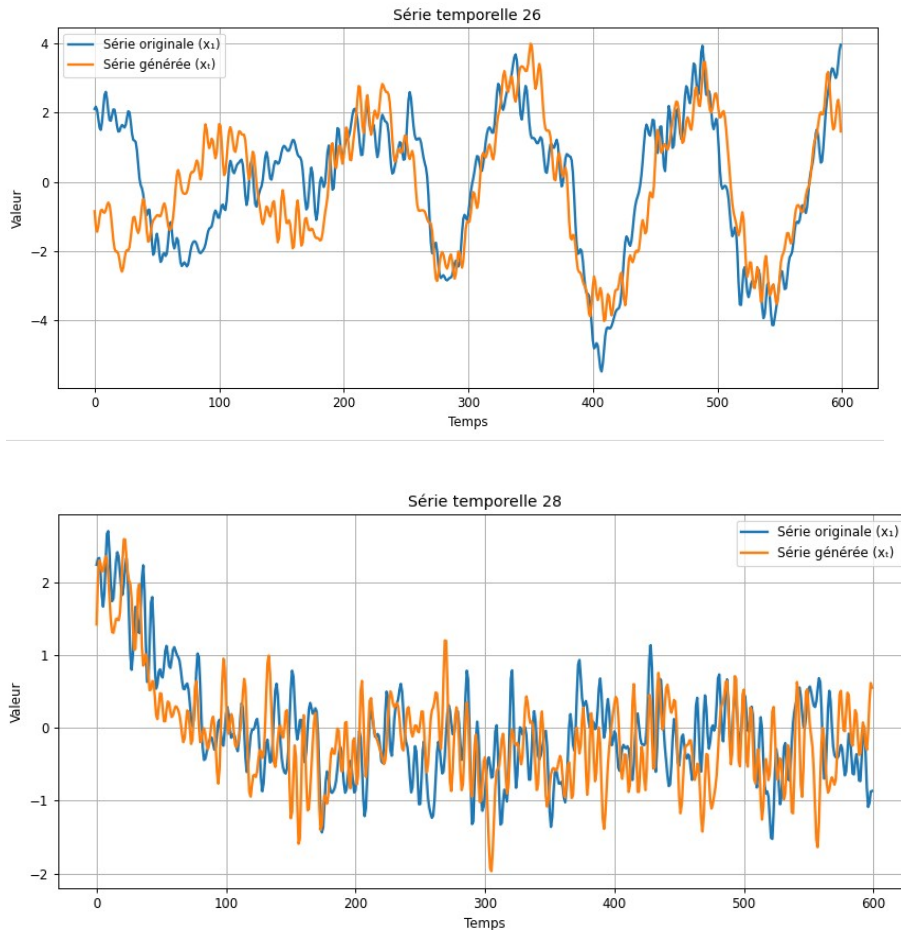
Data is sourced from the open OMEGA database. Training was conducted on a dataset of 4.5 million samples. Time series preprocessing includes:

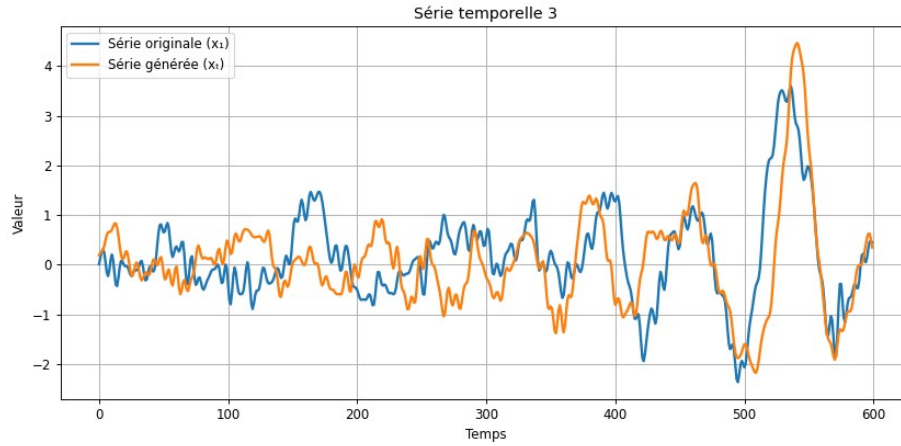
- Band-pass filtering between 0.5 Hz and 200 Hz.
- Downsampling to 1200 Hz.

Real-world inference is performed by extracting time windows containing the artifact (*on-event*), as well as adjacent windows (*pre-event* and *post-event*). The *catch22* features are computed on these segments and combined via a weighted average to form the target conditioning vector.

The ODE resolution during inference is performed using the Euler method with a constant integration step dt . The temporal grid follows a power-3 law, allocating higher integration density during the final denoising steps. A Classifier-Free Guidance (CFG) mechanism is applied with a scale factor between 2 and 5. Finally, the generated signal undergoes low-pass filtering via a Kaiser window ($\beta = 8.6$) to remove any potential high-frequency artifacts above 300 Hz.

Generated series example:





5. Preliminary Results

The evaluation of dynamic consistency relies on the Euclidean distance between the feature vectors of the original signal and those of the generated signal. Increasing the data volume from 1.5 million to 4.5 million samples reduced this average distance from 4.592 to 3.893.

When applied to ocular artifacts (blinks), the model was compared to reference methods (ICA, SSP). The evaluation metric used is the Cumulative Feature Distance, which measures the Euclidean deviation between the corrected segment and the adjacent clean segments. On a restricted sample ($n=4$ sessions), the generative model consistently demonstrates a lower distance (ranging from 9.69 to 9.86) compared to ICA (10.79 to 11.27) and SSP (10.94 to 11.15).

6. Limitations and Future Work

- **Statistical Validation:** Current results are based on a restricted sample (4 runs). Robust statistical validation on a larger cohort is required.
- **Spatial Topography:** The current pipeline operates in a single-channel manner, ignoring the spatial correlations and topographic structure inherent to magnetoencephalography recordings.
- **Theoretical Justification of the Sampler:** The choice of a constant-step Euler solver coupled with a power-3 law was driven by an empirical improvement in stability. A mathematical analysis of the ODE's local truncation error on this specific architecture would formally justify these inference parameters.
- **Scaling Constraints:** The utilization of the dataset was restricted by hardware memory limits. Scaling up to the entirety of the OMEGA database remains to be done.